

MODULE I: DEVELOP A MIXED-SIGNAL SYSTEM

4. Fmax:

Slow 1200mV 85C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	562.11 MHz	450.05 MHz	b2v_ins...[clk[1]	limit due to minimum period restriction (tmin)
2	615.01 MHz	450.05 MHz	b2v_ins...[clk[0]	limit due to minimum period restriction (tmin)

5. Logic %:

Flow Summary	
<<Filter>>	
Flow Status	Successful - Sun Sep 20 10:12:10 2026
Quartus Prime Version	16.1.0 Build 196 10/24/2016 SJ Lite Edition
Revision Name	pwm_led_top
Top-level Entity Name	pwm_led_top
Family	MAX 10
Device	10M50DAF484C6GES
Timing Models	Preliminary
Total logic elements	33 / 49,760 (< 1 %)
Total registers	31
Total pins	6 / 360 (2 %)
Total virtual pins	0
Total memory bits	0 / 1,677,312 (0 %)
Embedded Multiplier 9-bit elements	0 / 288 (0 %)
Total PLLs	1 / 4 (25 %)
UFM blocks	0 / 1 (0 %)
ADC blocks	0 / 2 (0 %)

6. LED0 starts dim, then using SW0,1, and 2, we can increase the duty cycle in 12.5% increments using binary. This is expected behavior!

9. Fmax:

Slow 1200mV 85C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	50.31 MHz	50.31 MHz	ADC_CLK_10	
2	62.64 MHz	62.64 MHz	altera_...ved_tck	
3	468.16 MHz	450.05 MHz	inst[a...][clk[0]	limit due to minimum period restriction (tmin)
4	473.71 MHz	450.05 MHz	inst[a...][clk[1]	limit due to minimum period restriction (tmin)

10: Logic %:

Flow Summary

 <<Filter>>

Flow Status	Successful - Sun Sep 20 11:10:30 2026
Quartus Prime Version	16.1.0 Build 196 10/24/2016 SJ Lite Edition
Revision Name	pwm_led_top
Top-level Entity Name	pwm_led_top
Family	MAX 10
Device	10M50DAF484C6GES
Timing Models	Preliminary
Total logic elements	4,751 / 49,760 (10 %)
Total registers	3233
Total pins	50 / 360 (14 %)
Total virtual pins	0
Total memory bits	270,176 / 1,677,312 (16 %)
Embedded Multiplier 9-bit elements	0 / 288 (0 %)
Total PLLs	2 / 4 (50 %)
UFM blocks	0 / 1 (0 %)
ADC blocks	1 / 2 (50 %)

11. The 7 segment display shows the ADC reading. With the jumper wire in place, the reading is relatively stable. On some values, like for duty cycle 12.5%, the 7 segment display oscillates between 158 and 159, but is otherwise stable. Adjusting the duty cycle also changes the ADC display accordingly. To increase the performance, the clock speed of the ADC could be increased. Then, more samples are taken.